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Public Key Encryption

Tasks:[https://seedsecuritylabs.org/Labs_20.04/Files/Crypto_PKI/Crypto_PKI.pdf]

Task 1:

What part of the certificate indicates this is a CA's certificate?
CA:TRUE.

```
            X509v3 Basic Constraints: critical
                CA:TRUE
    Signature Algorithm: sha256WithRSAEncryption
         0b:6f:de:25:90:40:aa:ad:27:74:72:6c:11:11:4c:ad:89:e9
         d9:73:18:92:b6:cd:f1:07:64:8d:bd:bf:7f:06:b4:ae:cb:24
         b4:d6:f8:7c:0e:e3:af:8a:f9:af:a6:77:9f:7b:0a:9a:f9:14
         c3:73:6a:ae:3f:4d:51:c3:90:f8:1d:1b:79:f4:e1:b6:95:05
```

What part of the certificate indicates this is a self-signed certificate?
The issuer and the subject are the same, which implies that the certificate is self-signed:

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            09:3b:90:df:ef:41:a5:2c:e9:ed:93:a0:ad:e5:cd:bb:9e:f5:2c:
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = www.nguyen2024.com, O = nguyen2024 Inc., C = CA
        Validity
            Not Before: Oct 13 19:23:47 2024 GMT
            Not After : Oct 11 19:23:47 2034 GMT
        Subject: CN = www.nguyen2024.com, O = nguyen2024 Inc., C = CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
```

In the RSA algorithm, we have a public exponent *e*, a private exponent *d*, a modulus *n*, and two secret numbers *p* and *q*, such that *n=pq*. Please identify the values for these elements in your certificate and key files:

modulus *n:*

```
Public Key Algorithm: rsaEncryption
    RSA Public-Key: (4096 bit)
    Modulus:
        00:e5:05:67:1f:de:75:71:71:1a:43:c6:9d:ce:1a:
        6a:b2:8b:60:ad:48:71:61:11:2d:5b:ac:9a:ff:ec:
        0b:a4:0d:0f:99:06:40:8d:76:ad:8a:2e:3d:16:e0:
        07:c8:fa:5a:12:2d:31:1f:c4:95:cc:1d:f0:71:f7:
        d2:47:76:f8:e4:49:e4:bb:fd:a2:9a:76:cd:89:c0:
        8b:d1:53:2a:c6:62:66:a7:10:2e:8f:63:5a:f5:1c:
        ea:d0:67:05:eb:1b:57:f2:c5:6f:48:d6:76:cb:e9:
        d7:65:02:3b:20:88:aa:24:cb:83:f1:39:3c:78:05:
        2c:28:68:90:6b:ed:23:af:3e:14:47:ad:39:25:49:
        40:15:ac:82:e9:4e:c7:7d:54:e0:24:d0:24:1f:5b:
        13:f7:97:1d:51:7f:30:cc:94:ff:89:27:d0:e7:43:
        00:60:f1:ed:c7:5f:86:d4:9e:c7:54:8c:4a:6e:6a:
        8f:e8:1a:2c:96:6e:02:48:38:45:fb:7b:51:14:15:
        07:d0:26:11:e4:33:7b:80:e2:5a:94:cd:f8:82:d9:
        ed:88:76:64:90:56:2e:1e:3d:5c:3b:f3:c8:34:60:
        82:86:cb:ba:62:da:46:74:d3:0d:2c:5f:9a:62:e5:
        a9:b1:6d:c7:2f:39:83:cf:a1:6a:d8:bf:27:fb:6e:
        22:dc:68:24:b1:33:ec:6e:f3:99:8c:b7:f0:ef:fb:
        dd:f8:a1:4d:aa:8e:cd:94:14:89:27:c4:a1:b8:df:
        5e:55:54:7a:b0:8a:5e:50:06:f0:11:77:91:df:03:
        c2:f7:e3:64:a2:3d:61:1a:5d:46:b4:1d:01:ae:eb:
        27:89:4f:90:ae:ef:15:f1:a5:87:39:c3:f4:8e:61:
        c3:09:9d:c9:fe:e6:bb:c5:90:ce:ff:88:f9:34:03:
        c2:b4:44:e8:51:ea:5b:ae:55:37:c5:bf:f2:79:10:
        41:b6:32:42:4b:55:8b:98:4b:1d:a4:ba:30:36:bd:
        8e:d5:ec:5b:e7:8e:b6:a3:82:e2:b6:09:d0:ec:f7:
        5c:d8:8f:58:aa:50:d1:33:0a:e9:9f:9c:ca:c2:dd:
        9b:68:08:d0:52:69:f7:bd:a0:34:8d:b5:da:eb:69:
        d4:c7:c7:f0:b9:cd:63:6d:2e:5e:2d:56:0c:7e:7d:
        22:f1:23:a8:42:5f:63:f1:5d:77:50:e4:0c:b8:45:
        ab:9c:94:24:37:50:fb:f2:07:52:a2:ba:61:75:e1:
        91:c7:39:48:a1:e8:3e:20:39:3b:fa:bc:69:8d:12:
        b8:70:68:8f:0f:4d:56:e7:14:3b:c5:20:ad:00:7a:
        6e:4b:60:57:11:32:6e:32:23:9e:7e:ba:99:9a:67:
        a1:2b:9d
    Exponent: 65537 (0x10001)
```

public exponent *e* and private exponent *d:*

```
publicExponent: 65537 (0x10001)
privateExponent:
    00:cd:3a:f7:3b:d6:d5:75:d0:3e:12:28:65:77:b1:
    b2:95:c7:52:de:56:01:cc:53:61:4b:3a:b5:37:cb:
    e6:6e:9f:89:af:be:ab:90:59:9e:90:2a:79:72:0b:
    56:ce:a2:92:9e:a1:42:d9:4e:08:52:ff:2d:5e:64:
    0f:5b:1d:61:52:ea:69:00:91:fc:e9:e0:b6:b5:d0:
    07:3b:91:04:ce:fc:95:32:f6:cc:52:90:5d:8d:33:
    d9:b4:e1:ed:aa:cf:39:14:04:1b:82:b1:35:89:ec:
    46:aa:6c:c2:61:98:36:a8:7c:7c:03:6a:20:27:09:
    b3:95:8e:86:85:40:df:ee:85:7a:14:f5:45:35:85:
    9b:d9:10:5b:cc:09:5d:b9:91:9a:bf:42:8a:6c:52:
    e8:e4:c1:d5:73:17:c0:ba:44:77:6e:08:71:2b:1c:
    9e:b5:41:77:ac:67:8d:62:f1:da:7c:88:af:96:9d:
    05:d2:9f:c9:ff:10:1c:19:6d:dc:40:cd:34:4e:03:
    be:99:a7:d3:f6:a4:af:f3:42:64:9b:de:ab:e9:4e:
    b3:8e:cb:51:b9:0a:91:8f:65:bb:46:ba:62:59:95:
    76:c2:a6:c4:a3:1b:29:72:3d:43:1b:d5:9c:7b:2d:
    85:d0:cb:28:68:c8:68:b7:62:d6:65:df:7f:4b:d0:
    59:89:69:b8:aa:96:43:fd:dd:62:85:e7:fb:9c:a9:
    13:35:3f:03:f3:72:eb:bd:e9:dd:20:f0:b2:3c:b6:
    23:ba:8d:15:8c:55:af:c8:f3:3f:da:52:ae:3a:6a:
    89:cf:fe:e5:71:a2:71:cd:0b:2b:a2:b4:c5:ad:bd:
    d1:52:e1:e5:a9:d8:6b:a1:7a:14:01:21:72:e0:95:
    24:44:56:7d:0b:99:c5:4c:78:19:2f:11:83:d4:1c:
    34:78:62:72:d8:ec:00:a4:18:9b:a2:3a:d3:d8:14:
    7a:2c:fd:da:1a:fe:74:02:be:21:31:98:8f:26:cb:
    d7:7a:92:7e:84:fa:ef:7d:b2:a4:07:e6:76:2f:3d:
    c3:89:e4:a3:02:85:0d:f0:cd:ab:13:8a:4f:f7:79:
    36:9d:7b:bf:30:21:92:ab:c3:4b:2f:8d:df:7a:59:
    a9:12:9d:85:d7:99:df:99:3b:3d:dd:dc:80:af:6c:
    41:53:9e:43:ce:6e:70:b0:80:a7:5a:c6:a3:22:d8:
    0d:e1:ce:8b:9c:68:b4:93:f9:b6:3b:59:62:0d:47:
    f4:b2:13:50:6a:d0:60:a4:32:e1:68:53:34:f3:7c:
    8c:55:8e:f6:7c:fb:9c:fa:26:7d:bc:14:1d:f1:52:
    af:8e:42:45:58:74:ee:34:40:bf:af:3e:0b:ae:a5:
    ac:dd:5d
```

secret numbers *p* and *q:*

```
prime1:
    00:fb:ca:a6:2a:a2:89:e9:07:c8:22:fa:65:6e:d6:
    f1:4e:1a:5f:05:06:3d:57:72:7f:4f:15:62:16:e1:
    81:57:43:0d:27:7a:b5:19:03:07:b7:91:ea:44:29:
    d1:d8:f8:c0:23:c6:c7:41:d4:28:ab:de:a3:aa:ae:
    21:9c:60:82:a0:11:36:c8:61:0b:cc:38:7f:28:2e:
    b1:7a:e4:2f:63:ce:81:ac:72:0c:c0:13:63:e2:e4:
    f6:16:0b:d8:57:cf:94:45:d0:4a:1f:1b:16:72:c7:
    ba:f7:f7:94:e1:df:40:a4:a0:66:a2:2d:40:cd:47:
    21:94:d7:9b:59:48:13:f9:98:04:4e:9f:c4:12:15:
    ba:48:6b:d8:16:17:a7:b2:a1:ef:8f:37:f5:ae:af:
    41:42:c6:71:0b:9e:7a:ff:53:2f:2b:67:e7:22:e0:
    e9:33:4d:a8:ad:e7:10:60:9f:04:a3:fa:87:09:a2:
    b8:98:a4:ae:63:9a:7a:42:af:20:15:d4:54:c0:c3:
    02:f7:7e:bb:13:46:ee:61:9b:fb:5d:e0:a8:92:d0:
    0d:d3:08:d9:44:ce:2e:7b:c9:b5:12:41:f9:3b:95:
    93:0e:78:3b:52:80:2f:03:e3:ad:42:81:ae:84:ee:
    3f:b1:de:56:0c:85:51:0a:41:94:2b:d3:b6:4d:2c:
    ef:d3
prime2:
    00:e8:d9:53:20:34:ad:c8:ef:26:4b:a5:8d:6d:df:
    cc:ef:66:6c:96:b7:21:fc:d7:d7:5e:00:05:56:7e:
    4c:64:f6:12:96:7c:71:34:71:41:dd:91:a4:6d:12:
    93:37:b5:a5:f8:ee:20:f4:fd:45:43:b9:1d:10:a0:
    0a:4d:8e:3b:99:85:40:54:42:de:9f:85:79:d3:05:
    7e:45:1c:1f:97:6d:7a:dc:99:e8:88:9f:e7:a8:7c:
    41:6b:82:ec:00:58:02:e1:db:b0:ba:48:d7:80:1e:
    12:f5:3f:c7:ee:f2:23:4e:8a:cf:18:e0:31:14:e3:
    e2:c3:f0:1b:5d:34:ab:9b:93:1a:8b:b3:46:ff:c6:
    f3:b2:91:18:0f:74:8e:12:28:41:07:08:60:bc:1d:
    56:fc:a9:a7:e5:67:92:e2:f7:aa:90:e3:2c:45:3a:
    9d:6a:4e:00:17:f6:16:57:51:05:00:a0:33:85:db:
    93:d3:a4:51:54:34:5a:48:08:90:03:d5:01:75:4a:
    9e:03:3b:d9:71:a3:ec:89:f9:ea:d3:cc:73:1a:34:
    65:7b:f2:35:d8:70:d8:16:e0:9f:b1:60:53:09:88:
    0b:a0:35:3a:5d:1b:bb:56:fb:77:5c:78:a6:be:45:
    a9:4e:ea:4c:21:b9:fd:19:fd:ad:27:51:14:6a:bf:
    c0:cf
```

Task 2:

First few lines of server.csr:

```
root@b00e913285f3:/# openssl req -in server.csr -text -noout
Certificate Request:
    Data:
        Version: 1 (0x0)
        Subject: CN = www.nguyen2024.com, O = nguyen2024 Inc., C = CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                RSA Public-Key: (2048 bit)
                Modulus:
                    00:c0:70:0c:5c:eb:75:7a:03:03:36:1d:a0:4a:f0:
                    2c:2f:87:6b:34:c1:2f:c3:1a:c0:5a:ad:d3:1c:25:
                    55:71:37:69:b5:3e:48:44:3b:36:62:7f:eb:60:0c:
                    65:55:8c:aa:dd:8e:3a:8d:b9:ab:1d:a9:a0:0c:b7:
                    6e:c2:40:c2:19:de:e2:95:02:4b:bf:9e:1d:b8:86:
                    81:a0:e7:3c:67:01:45:c4:7b:7a:2d:fd:f7:a4:b9:
                    a7:f9:26:58:f6:10:3b:96:18:73:29:cc:4e:8b:d8:
                    16:f5:3a:6a:d2:dd:c3:41:61:33:21:4a:8e:4b:9f:
                    8d:5a:51:eb:96:d9:f0:f9:bf:23:0a:fc:5f:c2:68:
                    b8:fa:7c:11:24:35:02:30:9b:fe:5e:57:f0:93:02:
                    bc:1f:0e:99:1e:9c:de:05:49:bc:f6:64:bd:0b:f7:
                    dc:df:59:98:73:6c:b5:2e:ed:16:10:3d:fb:f2:c0:
                    9e:46:94:eb:90:bb:2f:fd:07:37:9c:eb:cb:65:c9:
                    3b:d1:9e:53:a2:4a:c6:d8:0e:9b:37:fc:35:44:6e:
```

Adding alternate names:

```
root@b00e913285f3:/# openssl req -newkey rsa:2048 -sha256  \-keyout server.key
 -out server.csr  \-subj "/CN=www.nguyen2024.com/O=nguyen2024 Inc./C=CA" \-passo
ut pass:dees -addext "subjectAltName = DNS:www.nguyen2024.com,  \DNS:www.nguyen2
024A.com, \DNS:www.nguyen2024B.com"
Generating a RSA private key
.....................................+++++
.....................................................................+++++
writing new private key to 'server.key'
```

Task 3:

Generating Certificate:

```
root@b00e913285f3:/demoCA# openssl ca -config openssl.cnf -policy policy_anything \-md sha256 -days 3650 \-in server.csr -out
ch \-cert ca.crt -keyfile ca.key
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
        Serial Number: 4096 (0x1000)
        Validity
            Not Before: Oct 13 22:25:48 2024 GMT
            Not After : Oct 11 22:25:48 2034 GMT
        Subject:
            countryName               = CA
            organizationName          = nguyen2024 Inc.
            commonName                = www.nguyen2024.com
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            Netscape Comment:
                OpenSSL Generated Certificate
            X509v3 Subject Key Identifier:
                F9:E4:E2:E6:31:D2:38:66:BB:EC:D3:CD:22:00:26:94:98:97:1C:B1
            X509v3 Authority Key Identifier:
                keyid:69:15:30:02:0D:49:FB:B1:7F:95:1C:71:C8:AE:BE:37:52:19:B2:7D

            X509v3 Subject Alternative Name:
                DNS:www.nguyen2024.com, DNS:www.nguyen2024A.com, DNS:www.nguyen2024B.com
Certificate is to be certified until Oct 11 22:25:48 2034 GMT (3650 days)

Write out database with 1 new entries
Data Base Updated
```

Decoded Content (alternative names are included):

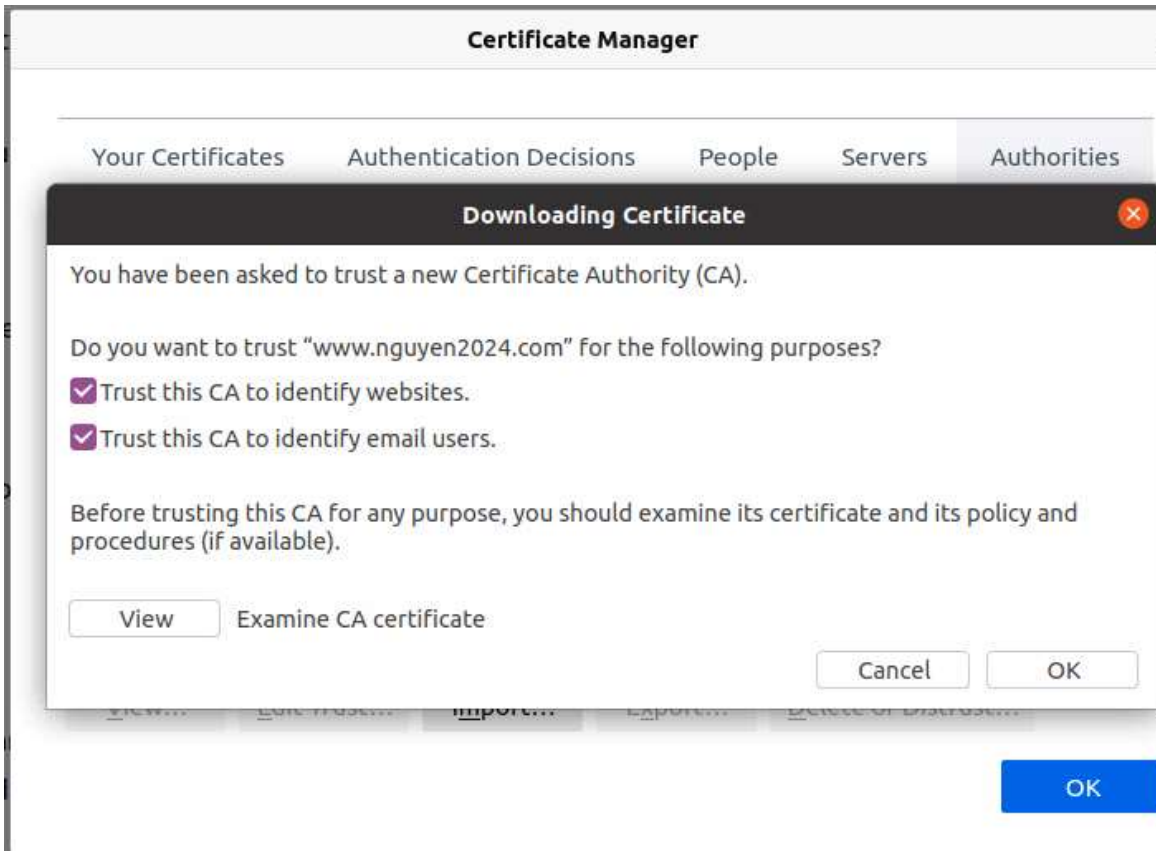
```
root@b00e913285f3:/demoCA# openssl x509 -in server.crt -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 4096 (0x1000)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = www.nguyen2024.com, O = nguyen2024 Inc., C = CA
        Validity
            Not Before: Oct 13 22:25:48 2024 GMT
            Not After : Oct 11 22:25:48 2034 GMT
        Subject: C = CA, O = nguyen2024 Inc., CN = www.nguyen2024.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                RSA Public-Key: (2048 bit)
                Modulus:
                    00:a8:8e:e0:d4:e5:24:42:2a:77:a9:e1:ea:d0:51:
                    27:7e:86:70:3e:31:9e:67:a0:9a:b3:f2:3e:5b:41:
                    d7:ea:5c:c7:56:7f:37:6e:d1:fc:be:82:52:56:ae:
                    05:a9:6c:5e:59:47:05:f6:bb:2d:96:f8:2a:dd:3a:
                    e8:62:b0:ff:e0:4a:36:d5:5e:80:1c:c3:48:f3:cf:
                    2c:d6:14:3c:0e:a1:57:06:90:10:23:81:69:bc:b5:
                    06:8e:bb:f0:d2:e5:dc:b2:0a:1f:ab:65:8a:26:f3:
                    40:1a:29:0b:13:eb:5a:d1:cb:7e:30:8b:19:00:b8:
                    5e:3e:0c:12:3f:ac:d6:b9:12:f6:28:ee:7f:41:03:
                    45:50:e7:a1:df:28:d7:53:53:9a:e2:0d:f1:b4:9c:
                    63:68:c8:44:e8:2e:70:d5:e4:e5:0a:ff:73:0c:74:
                    7c:72:d6:e5:81:49:d7:8e:99:b4:13:de:8f:6c:23:
                    01:74:c7:17:e1:44:b7:ea:e0:00:fc:20:99:70:34:
                    15:cd:96:9a:67:48:6d:db:07:ab:79:e6:92:ff:2a:
                    46:94:ef:71:f2:f0:a6:e1:56:f9:c6:6a:9e:6c:6d:
                    3a:7b:9b:b7:64:24:96:86:3c:11:68:73:a8:21:cc:
                    44:3a:4a:e4:7a:19:35:58:c5:05:b9:c5:3a:a9:52:
                    87:69
                Exponent: 65537 (0x10001)
```

```
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            Netscape Comment:
                OpenSSL Generated Certificate
            X509v3 Subject Key Identifier:
                F9:E4:E2:E6:31:D2:38:66:BB:EC:D3:CD:22:00:26:94:98:97:1C:B1
            X509v3 Authority Key Identifier:
                keyid:69:15:30:02:0D:49:FB:B1:7F:95:1C:71:C8:AE:BE:37:52:19:B2:7D

            X509v3 Subject Alternative Name:
                DNS:www.nguyen2024.com, DNS:www.nguyen2024A.com, DNS:www.nguyen2024B.com
    Signature Algorithm: sha256WithRSAEncryption
         44:22:f2:03:0b:91:b2:d4:d4:a9:22:c3:48:12:23:4b:6b:65:
         0a:bf:f8:12:05:4d:73:62:ad:a3:4e:7a:bf:54:d4:7b:e7:c1:
         03:1e:97:cc:f9:71:a3:19:01:79:7b:b7:41:17:4b:0a:de:98:
         27:b2:18:54:ad:8e:e2:9c:63:4f:67:df:e7:e4:d1:9c:75:55:
         ba:87:d8:79:e6:90:2a:0d:9b:fe:25:1a:97:31:86:09:51:4c:
         57:8a:1c:2c:6f:56:5f:0e:bd:b3:c2:c7:bb:41:42:54:52:5d:
         66:1a:31:e0:d7:eb:e2:0b:73:ce:94:d4:27:a4:fa:ca:f2:8e:
         c1:92:25:f8:b1:d5:25:23:e9:ef:db:77:8a:1a:fe:ac:3b:60:
         4e:cd:8a:d7:1b:48:ed:a9:a2:86:5a:12:56:b8:3f:79:74:50:
         c5:f5:8b:47:a5:6d:e3:f1:64:05:73:8c:aa:d3:6d:26:e0:28:
         c6:61:ce:f2:ee:d6:db:d4:09:ef:94:85:38:2a:ce:c6:53:c2:
         af:60:ec:fa:08:c9:cb:29:5f:28:c7:e0:93:16:21:d4:cc:20:
         5c:f2:98:2a:5f:8e:9a:3d:19:66:cf:e7:5a:c8:8c:13:33:cc:
         b7:ec:ce:d9:86:7e:5b:31:c8:82:3c:4e:d5:c1:0b:56:9e:7a:
         16:28:a3:5e:fa:68:79:22:03:d9:4a:33:9d:6e:9b:10:42:f8:
         84:e3:68:cd:72:fe:68:76:75:e5:fd:52:36:55:9a:2e:3b:8d:
         48:5d:9a:35:50:9c:c1:fe:83:3e:61:82:5d:77:36:ab:0e:c5:
         5d:87:b8:46:07:97:63:40:9a:7b:fd:94:38:3a:6a:98:26:9d:
         31:7b:71:79:54:99:65:e7:e4:c8:dd:cd:16:9a:82:fb:1f:e6:
         c6:2d:73:56:af:bb:7a:b1:26:93:07:b6:53:f5:72:25:1d:b1:
```

^ www.nguyen2024A.com and www.nguyen2024B.com are included in the SAN section

Task 4:



Task 5:

ahmed2024_apache_ssl.conf:

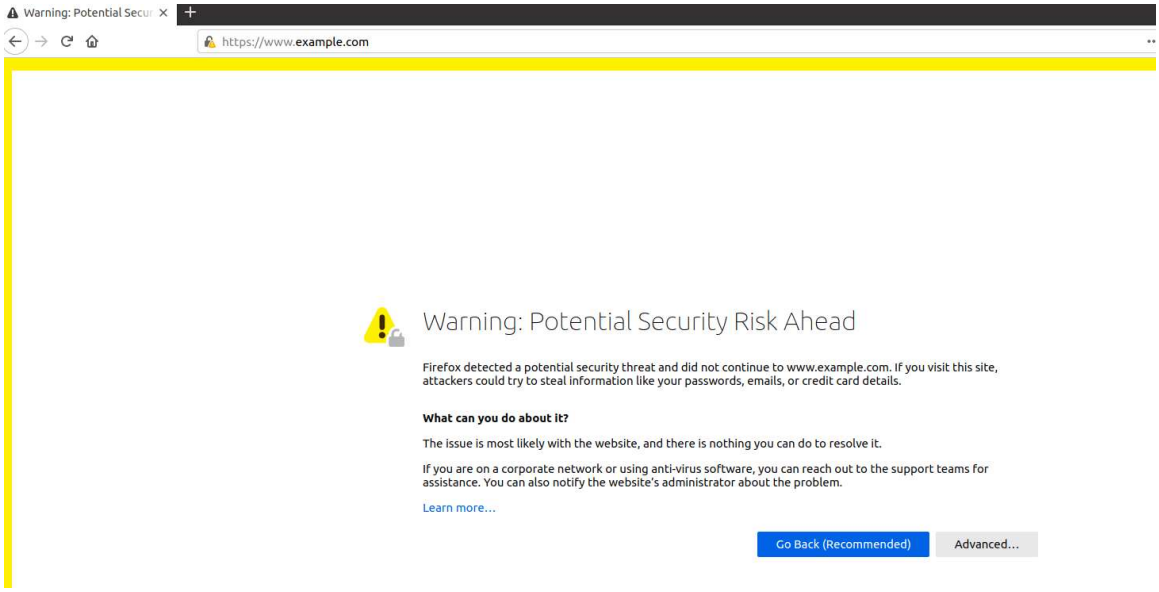
```
<VirtualHost *:443>
    DocumentRoot /var/www/ahmed2024
    ServerName www.ahmed2024.com
    ServerAlias www.ahmed2024A.com
    ServerAlias www.ahmed2024B.com
    ServerAlias www.ahmed2024W.com
    DirectoryIndex index.html
    SSLEngine On
    SSLCertificateFile /certs/ahmed2024.crt
    SSLCertificateKeyFile /certs/ahmed2024.key
</VirtualHost>

<VirtualHost *:443>
    DocumentRoot /var/www/example
    ServerName www.example.com
    ServerAlias www.exampleA.com
    ServerAlias www.exampleB.com
    ServerAlias www.exampleW.com
    DirectoryIndex index.html
    SSLEngine On
    SSLCertificateFile /certs/ahmed2024.crt
    SSLCertificateKeyFile /certs/ahmed2024.key
</VirtualHost>
```

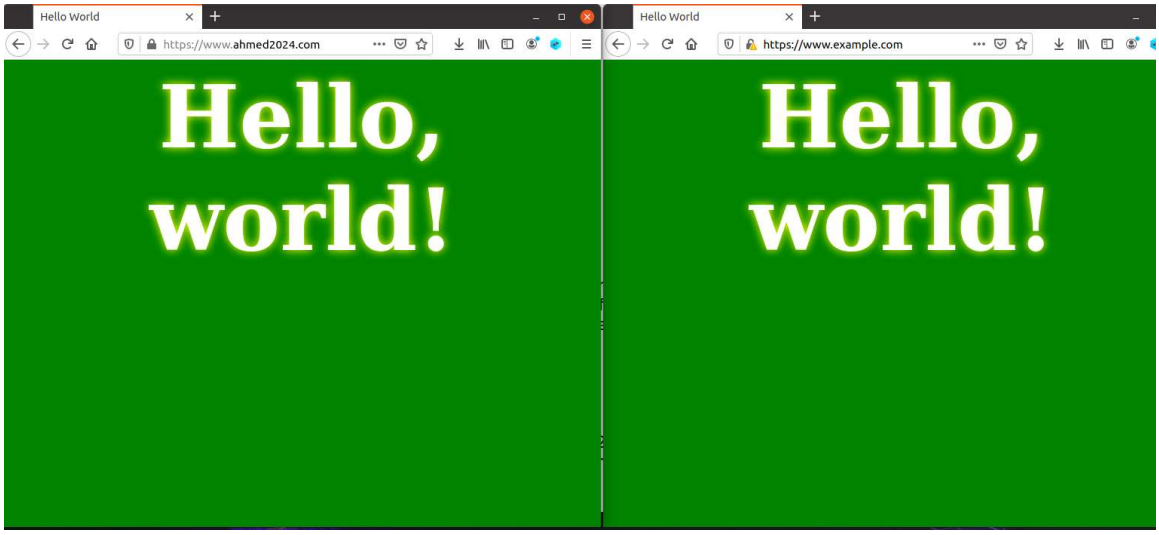
/etc/hosts:

```
10.9.0.80       www.bank32.com
10.9.0.80       www.ahmed2024.com
10.9.0.80       www.example.com
```

When accessing https://www.example.com, the first thing that appears is a warning:



This is expected since we are using a CA certificate that was copied from the original website. When ignoring the warning, the results when accessing the fake site were an identical webpage to www.ahmed2024.com. This demonstrates a man-in-the-middle attack scenario where someone could potentially steal user credentials by creating a fake log-in page. The user would think this is a real website and enter their information, not realizing that they are on a fake website.



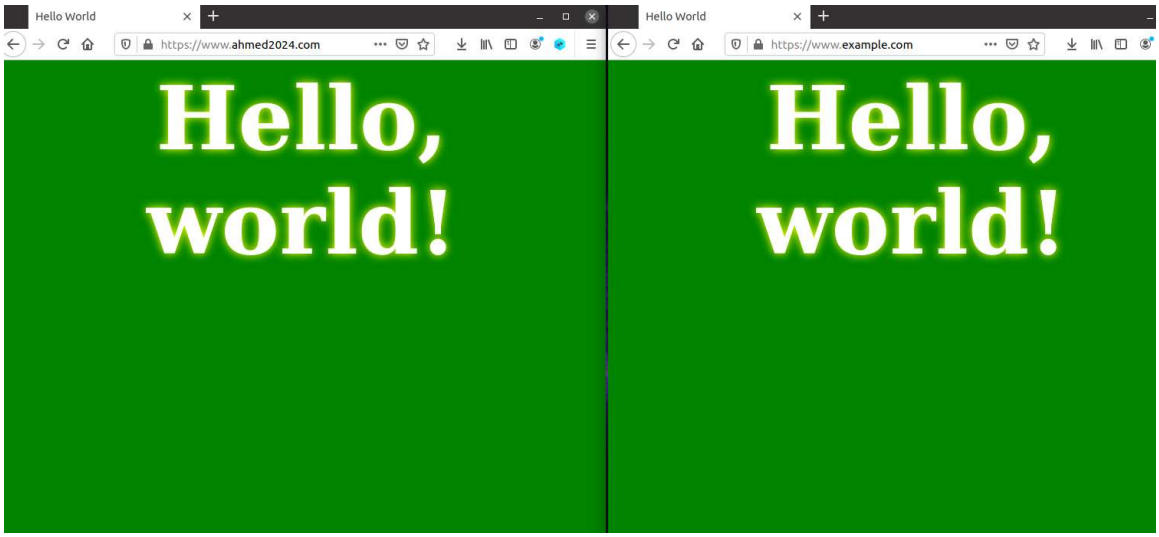
However, since the fake website is unsecured and displays a warning before accessing it, the user should realize that this is a fake website.

Task 6:

```
<VirtualHost *:443>
    DocumentRoot /var/www/ahmed2024
    ServerName www.ahmed2024.com
    ServerAlias www.ahmed2024A.com
    ServerAlias www.ahmed2024B.com
    ServerAlias www.ahmed2024W.com
    DirectoryIndex index.html
    SSLEngine On
    SSLCertificateFile /certs/ahmed2024.crt
    SSLCertificateKeyFile /certs/ahmed2024.key
</VirtualHost>

<VirtualHost *:443>
    DocumentRoot /var/www/example
    ServerName www.example.com
    ServerAlias www.exampleA.com
    ServerAlias www.exampleB.com
    ServerAlias www.exampleW.com
    DirectoryIndex index.html
    SSLEngine On
    SSLCertificateFile /certs/fake_example.crt
    SSLCertificateKeyFile /certs/ahmed2024.key
</VirtualHost>
```

By accessing the private key and making a new certificate, the fake website doesn't display a warning anymore. This is very dangerous:



As you can see, both websites are identical. The only way a user could tell that they are on a fake website is by looking at the URL. This is dangerous since an attacker could use a URL that is very similar to the real URL. For example, the attacker could use https://www. ahhmed2024.com and the user wouldn't notice. Say the user was trying to type in the website URL and mistyped it, he would land on the fake website without even noticing. The easiest way to avoid this would be to use bookmarks. This shows that

protecting your private key is a very important factor in ensuring that your website doesn’t get compromised.